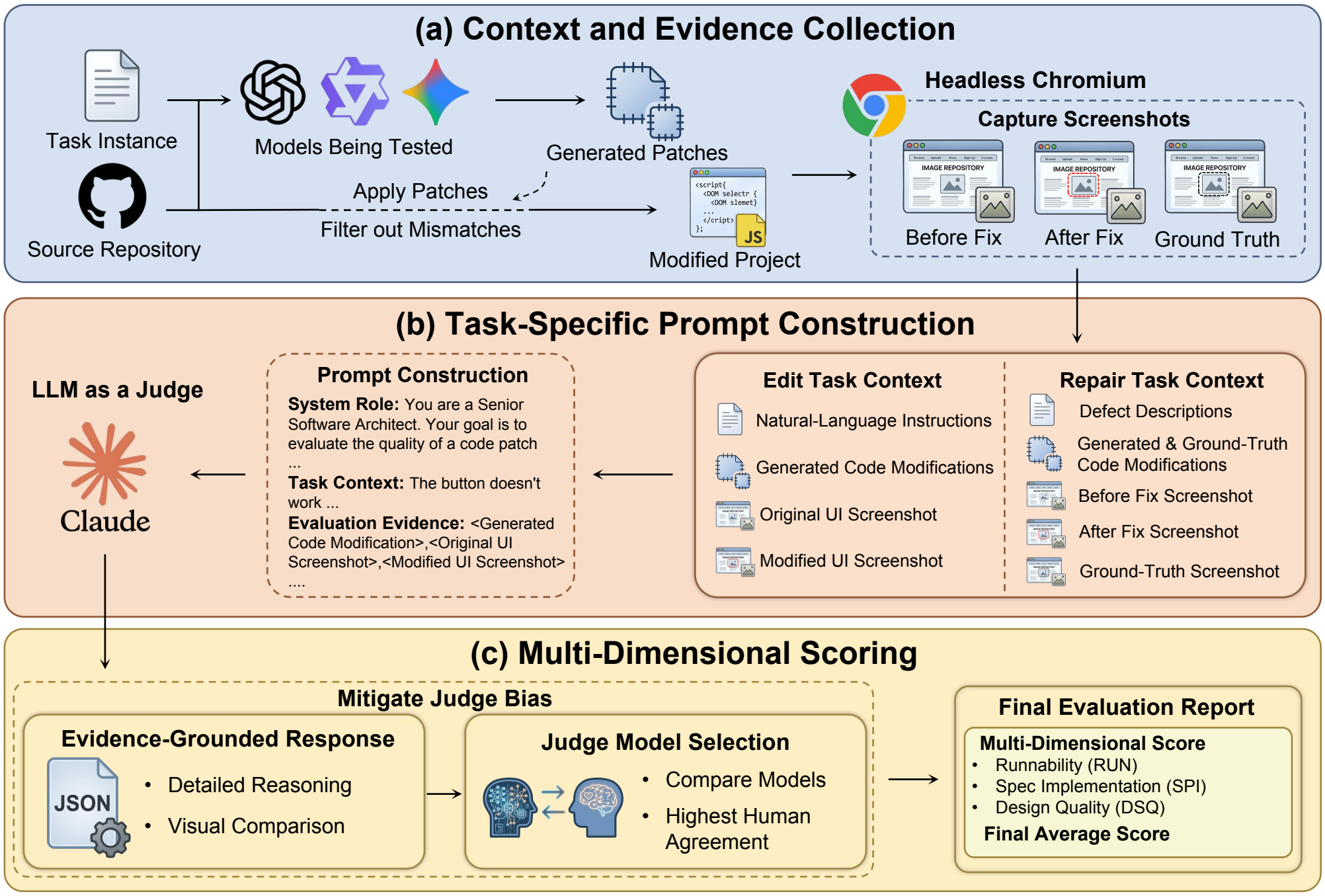
(a) Context and Evidence Collection
Task Instance
Source Repository
Models Being Tested
Generated Patches
Apply Patches
Filter out Mismatches
Modified Project
Headless Chromium
Capture Screenshots
Before Fix
After Fix
Ground Truth
(b) Task-Specific Prompt Construction
LLM as a Judge
Claude
Prompt Construction
System Role: You are a Senior Software Architect. Your goal is to evaluate the quality of a code patch ...
Task Context: The button doesn't work ...
Evaluation Evidence: <Generated Code Modification>,<Original UI Screenshot>,<Modified UI Screenshot> ....
Edit Task Context
Natural-Language Instructions
Generated Code Modifications
Original UI Screenshot
Modified UI Screenshot
Repair Task Context
Defect Descriptions
Generated & Ground-Truth Code Modifications
Before Fix Screenshot
After Fix Screenshot
Ground-Truth Screenshot
(c) Multi-Dimensional Scoring
Mitigate Judge Bias
Evidence-Grounded Response
JSON
Detailed Reasoning
Visual Comparison
Judge Model Selection
Compare Models
Highest Human Agreement
Final Evaluation Report
Multi-Dimensional Score
Runnability (RUN)
Spec Implementation (SPI)
Design Quality (DSQ)
Final Average Score